

A Look at the Top Three Network Monitoring Tools



This article talks about a few important aspects of network monitoring, and compares three leading tools that are of great importance to IT administrators.

In a well managed IT infrastructure, network monitoring acts as the eyes and ears of an organisation to spot problems before they appear. Systems administrators need complete visibility into their critical IT components such as servers, applications and networks. These tools can monitor a server crash, a failing application, or in some cases, highly utilised network bandwidth.

Features of network monitoring tools

A network monitoring tool is usually hosted on a standalone server and runs its client software on each machine to be managed or monitored. The tool usually runs its own copy of the database such as MySQL or Postgres, which stores all scripts, historic events and actions. In some modern tools, an agent is not required to be run on managed machines, making it an agent-less installation. Table 1 lists a stack of features with examples, which must be available by default in a network monitoring tool.

When it comes to monitoring large scale IT infrastructures, systems administrators need architecture with much-advanced features to make their life easy. Given below is a list of some important features.

Auto discovery: It is cumbersome for administrators to add each managed device manually. Modern monitoring tools span the entire network segment to enumerate devices and perform auto discovery of operating system, configuration and settings. This feature automatically helps admins to get a glimpse of their IT inventory.

Network traffic stats: Earlier, monitoring tools used to just look at the CPU, memory and disk utilisation. However, this is not enough and network bandwidth usage is a key factor to be aware of, especially when the managed machines are supposed to access the Internet. Besides, by monitoring network traffic, admins get an insight into the bandwidth usage of the Internet service provider's line, which helps them make a capacity planning decision.

Log monitoring: All operating systems create activity logs. For example, in case of Linux, SSH logs and bash logs are created, while for Windows, the application, system and security event logs are generated. A good tool must be capable of reading and parsing log files. This sounds easy but can be tricky, because the operating system opens log files and locks those that require tools to sneak into the file without tampering or corrupting it.